ECS 251 Discussion: How to Give a Talk

1/5
ECS 251
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Q/A Opportunity

Any questions on course format?

Course policies?

Readings?

etc.?

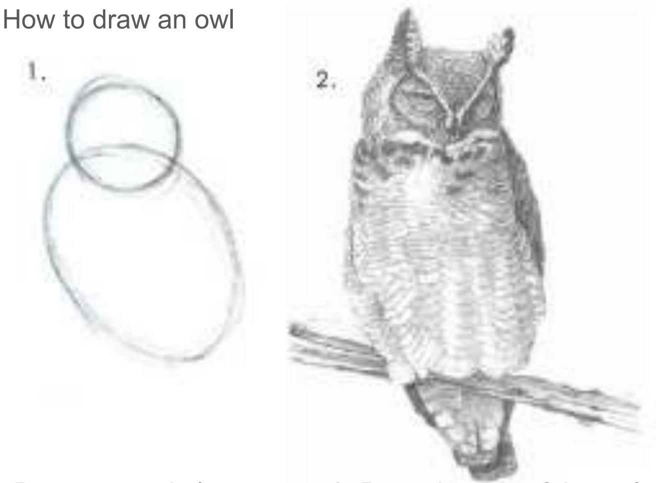
How to give a talk

Slides and advice adapted from Thorsten Hoefler (ETH Zurich) and Simon Peter (UWashington)

How to present a research paper

1. Read the paper (how to read a paper from class intro)
2. Present the paper (this discussion section)

How to draw an owl



1. Draw some circles 2. Draw the rest of the owl

Talking about research

- A good researcher can express their knowledge well
    - This assumes we have that knowledge!
    - So only advance to this step once you understand the paper :)
- Why is talking about research useful?
    - Order your thoughts, think about how to explain them
    - Communication with other researchers (your classmates!)
    - Gather feedback
    - Establish relationships
    - Eventually build a career

Why do you care?

- Presenting will be important for your career!
  - This is a good way to convince people to:
    - Give you good grades
    - Give you a job/promotion
    - Give you money/resources
    - Think you're smart
    - Like you, recommend you
- But presentation skills are hard to acquire
  - No one is good at this right away
  - Practice practice practice!

A good research talk

- Is centered around the audience (not you)
  - **Teaches**, engages, provokes, excites listeners
- Provides intuitions to the audience
  - Take away messages, surprises, “wow” effects
- It does not need to
  - Tell every detail (not possible anyway in the time limit)
  - Show off how smart you are

Tip: focus on clearly defined goals

- Pick your goals carefully
  - What do you want to communicate? What should people remember?

Anatomy of a talk

- Motivation, placement
  - 20%
- Key ideas
  - 70-80%
- Evaluation results
  - 0-10%
- Do not present results without an explanation
  - Don’t just say “the authors made Application A run 50% faster”
  - Instead say “the authors present the FOO method relying on intuition BAR and that achieves 50% improvement for Application A”

The beginning of your talk

- You have about 2 minutes before your audience dozes off or starts reading email
  - Use them! Make every second count
  - Good approach: Present an abstract of your talk (or an elevator pitch):
    - Problem / motivation
    - Approach / idea
    - Experiments / results
    - Broader meaning / impact
- Answer these questions within two minutes:
  - What is the problem?
  - Why is this talk interesting? Why should I listen?

Communicating the key idea

- Pick a goal for the talk
  - Plan and make key points in your head. Organize the whole talk around these key points. Pick no more than three (better: one)
  - Be explicit, very explicit: “If you remember nothing else from this talk, remember this”
  - Repeat, repeat (but don’t be annoying)
- Do NOT be shallow, be deep
  - Avoid overviews
  - Do NOT ramble
  - Get to the meat quickly (assume we’ve all read the paper already)

Examples

- Are your main weapon
- Make your own examples, try to avoid just using the paper’s
- Ideally have a motivating example at the beginning
  - Maybe pose a question to get the audience thinking:
    - e.g., “What is the maximum speedup if we can solve this problem?”
- Illustrate the idea in action
  - From different perspectives
  - Show corner cases, highlight shortcomings
- Images say more than 1000 words!

What to use

- Enthusiasm
  - Be excited, pull the audience with you
  - Don’t be afraid to move around, but don’t pace or fidget
- Your brain!
  - Review/polish your slides before the talk
  - Have the storyline down
  - Focus on the key ideas
- Animations and graphics
  - Can be helpful (aesthetically as well as informative)
  - Nice slides make people more receptive

What to omit

- Do not present excessive related work
  - You can mention it in your slides, or have backup slides
  - Give credit
- Do not present too many technicalities
  - Audience probably won’t follow
  - Put details in the backup slides in case you do get a question on them
- Do not exaggerate with animations
  - Animations are good, but too many are difficult to follow
- Do not clutter your slides

How to present

- Be (or at least appear) confident
  - Don’t forget to breathe! :)
- Make eye contact
  - Look around, don’t stare at a single person
  - Tip: identify a nodder (these people always exist). They’ll give you confirmation
- Watch the audience
  - Sometimes they ask questions, don’t let them interrupt you but serve their questions
  - Questions are great, ask some and answer them!
- Finish on time!
  - Skip slides if necessary, never ask “should I continue?”
    - No polite person would ever say, “no, thanks”

Miscellaneous

- Standard stuff
  - Aviod erorrs no sldies
  - Face the audience
  - Check your laptop before
- Practice, practice, practice
  - Give your talk a couple of times before you present it publicly
  - Present to your group mates, me during office hours, your friends, your cat…
- You’ll attend more talks here than you’ll give
  - Engage, help your fellow students!
  - Ask questions, participate in the discussion, be awake

Miscellaneous

- You may borrow slides
  - But, be aware of the context of the slides
    - E.g., conference slides are intended for a very specialized audience
  - May have to adapt the slides for a more general audience (and the time frame)
  - And give credit
- Timing and cadence
  - Between 1-2 minutes per slide is common
  - Try not to rush too quickly through slides or spend too much time on a slide

Suggested lecture structure

- 5 minutes on background
  - What problem is this paper trying to solve? What other solutions have been attempted? Why do they fall short? Any historical context?
- 10 minutes on core concepts
  - What is this paper’s solution to the problem? What is their core insight or design? Why is this paper important?
- 10 minutes for small group discussion
  - Your group and I will walk around the groups and help answer questions
- 15 minutes whole class discussion
  - Your group will help lead a full class discussion, perhaps based on questions from groups

Quiz questions to go with the lecture

- You will develop 4 quiz questions based on the reading
  - Should be on the important concepts or ideas from the paper, not minute details
    - E.g., **don’t** ask “How much faster was Exokernel than Ultrix for null procedure calls?”
    - **Do** ask “What are secure bindings in Exokernel?”
- Can be definition-based questions (“What is <thing>?”) or more open ended questions (“Why did the authors go with design decision A?” or “Compare/contrast” questions)
- We may use your quiz questions when developing the weekly quizzes!
  - Your opportunity for easy points! ;)

How your presentation will be graded (lead a lecture)

- Clarity of background (3 pts)
- Clarity of main concept (6 pts)
- Smoothness and polish of presentation (2 pts)
- Question and answer leading (2 pts)
- Turning in slides by the class before your lecture (3 pts)
- Turning in 4 potential quiz questions (4 pts)

How your presentation will be graded (final project)

- Clarity and quality of presentation (2 pts)
- Explanation of problem and context (3 pts)
- Explanation of the system you've built (6 pts)
- Handling of the Q/A session (3 pts)